

# MAC AND IP ADDRESSING: SAME NETWORK vs REMOTE NETWORK

## 9.1.1 DESTINATION ON SAME NETWORK

- A host uses the destination IP address to find the device.
- If the destination is on the same network, the host uses the **destination MAC address** to deliver the frame.

### KEY ADDRESSES



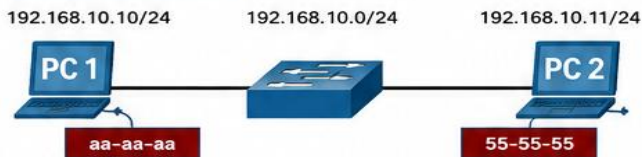
**Physical address (MAC)** – Used for NIC to NIC communication on the same Ethernet LAN.



**Logical address (IP)** – Used to send the packet from source to destination. The destination IP may be on the same or a remote network.

Layer 2 (MAC) addresses deliver the Ethernet frame between NICs on the same network. If the destination IP is on the same network, the frame is sent directly to the destination device.

### EXAMPLE TOPOLOGY



| Destination MAC | Source MAC | Source IPv4   | Destination IPv4 |
|-----------------|------------|---------------|------------------|
| 55-55-55        | aa-aa-aa   | 192.168.10.10 | 192.168.10.11    |

### WHAT HAPPENS?

- PC1 wants to send a packet to PC2.
- PC1 knows PC2 is on the same network.
- PC1 uses ARP to get PC2's MAC address (55-55-55).
- PC1 creates the Ethernet frame and IP packet.

### LAYER 2 (ETHERNET FRAME) CONTAINS:



- Destination MAC:** 55-55-55 (PC2)
- Source MAC:** aa-aa-aa (PC1)

### LAYER 3 (IP PACKET) CONTAINS:



- Source IP:** 192.168.10.10 (PC1)
- Destination IP:** 192.168.10.11 (PC2)



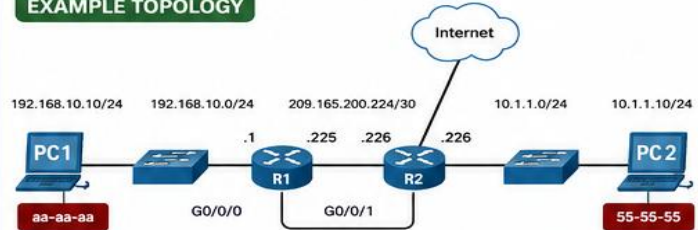
### SUMMARY

On the same network, the frame is sent directly to the destination device using its MAC address.

## 9.1.2 DESTINATION ON REMOTE NETWORK

- If the destination IP is on a remote network, the packet is sent to the default gateway (router interface on the local network).
- The router then forwards the packet to the remote network.

### EXAMPLE TOPOLOGY



| Destination MAC | Source MAC | Source IP     | Destination IP |
|-----------------|------------|---------------|----------------|
| (see below)     | aa-aa-aa   | 192.168.10.10 | 10.1.1.10      |

### STEP 1 – PC1 SENDS TO DEFAULT GATEWAY (R1)

- Destination MAC:** R1 G0/0/0 interface (bb-bb-bb)
- Source MAC:** PC1 (aa-aa-aa)
- Source IP:** 192.168.10.10
- Destination IP:** 10.1.1.10

### STEP 2 – R1 FORWARDS TO NEXT HOP (R2)

- Destination MAC:** R2 G0/0/1 interface (cc-cc-cc)
- Source MAC:** R1 G0/0/1 interface (bb-bb-bb)
- Source IP:** 192.168.10.10 (unchanged)
- Destination IP:** 10.1.1.10 (unchanged)

### STEP 3 – R2 SENDS TO DESTINATION (PC2)

- Destination MAC:** PC2 (55-55-55)
- Source MAC:** R2 G0/0/1 interface (dd-dd-dd)
- Source IP:** 192.168.10.10 (unchanged)
- Destination IP:** 10.1.1.10 (unchanged)

### RULES TO REMEMBER

- ✓ Hosts send to the default gateway when the destination is remote.
- ✓ Routers change the Layer 2 (MAC) addresses at every hop.
- ✓ Layer 3 (IP) addresses stay the same end-to-end.

## SUMMARY COMPARISON

| Feature                 | Same Network (Direct)                      | Remote Network (Via Router)             |
|-------------------------|--------------------------------------------|-----------------------------------------|
| Destination IP          | On the same IP network                     | On a different IP network               |
| Next Hop                | The destination device                     | The default gateway (router interface)  |
| Layer 2 Destination MAC | MAC of the destination device              | MAC of the default gateway              |
| Layer 3 IP Addresses    | Source IP and Destination IP in the packet | Same (unchanged) across the entire path |
| MAC Address Changes     | No (only one frame sent)                   | Yes, at every router hop                |
| Example                 | PC1 → PC2 on 192.168.10.0/24               | PC1 → R1 → R2 → PC2 on 10.1.1.0/24      |

### KEY TAKEAWAYS



MAC address is for delivery on the local link (Layer 2).



IP address is for end-to-end delivery (Layer 3).



Same network = send directly to destination MAC.



Remote network = send to default gateway first.



Routers change MAC addresses, not IP addresses.

# ARP (ADDRESS RESOLUTION PROTOCOL) – PART 1

## 9.2.1 ARP OVERVIEW

ARP (Address Resolution Protocol) maps IPv4 addresses to MAC addresses.

Every device on an Ethernet network has:

- IP** • **Logical address (IP)** – Used to send packets from source to destination. May be on the same network or a remote network.
- MAC** • **Physical address (MAC)** – Used for NIC to NIC communication on the same Ethernet LAN.

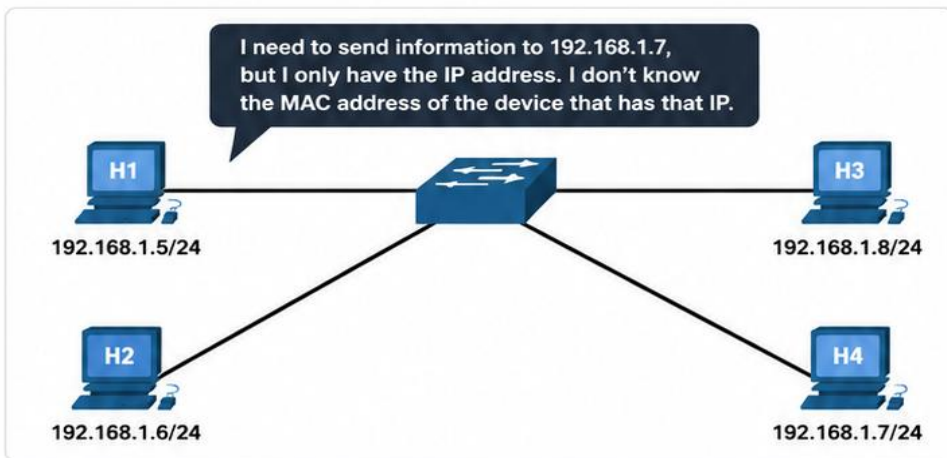


### KEY POINTS

- **Layer 2 (MAC)** addresses deliver Ethernet frames between NICs on the same network.
- If the destination IP is on the **same network**, the destination **MAC** is the MAC of the destination device.
- If the destination IP is on a **different network**, the destination **MAC** is the MAC of the **default gateway** (router interface).

### THE PROBLEM

A host knows the destination IP address, but not the destination MAC address. It needs the **MAC** address to send the Ethernet frame.



### ARP PURPOSE

ARP is used to discover the MAC address of a device when its IPv4 address is known.

ARP provides:

- Resolving IPv4 addresses to MAC addresses

## 9.2.2 ARP FUNCTIONS

Before sending a frame, the device checks its ARP table (stored in RAM) for a mapping of the destination IPv4 address to a MAC address. This table is called the ARP table or ARP cache.

### WHEN THE DEVICE CHECKS THE ARP TABLE



**Same Network:** If the destination IP is on the same network as the source IP, the device searches the ARP table for that destination IP.



**Different Network:** If the destination IP is on a different network, the device searches the ARP table for the IP of the default gateway.

### ARP TABLE (CACHE)

- Each entry maps an IPv4 address to a MAC address.
- Example format:  
192.168.1.7 → aa-bb-cc-dd-ee-ff
- The table is temporary and stored in RAM.

### AFTER THE SEARCH



**Entry Found:** Use the corresponding MAC address as the destination MAC in the frame.



**Entry Not Found:** Send an ARP request to discover the MAC address.



### IMPORTANT

In both cases, the device searches for an IPv4 address and a corresponding MAC address for the device.

### HOW ARP WORKS (SUMMARY PROCESS)



### KEY TAKEAWAY

ARP enables communication on an Ethernet network by translating IPv4 addresses to MAC addresses.

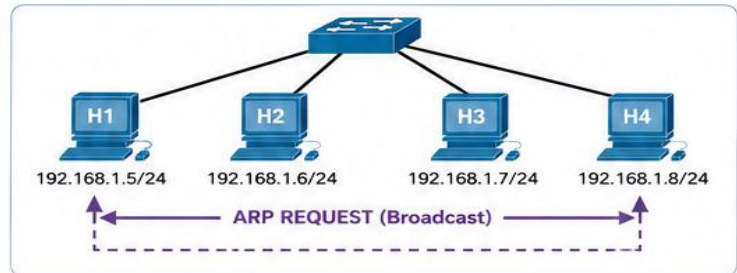
# ARP (ADDRESS RESOLUTION PROTOCOL) – PART 2

## 9.2.3 VIDEO – ARP REQUEST

An ARP request is sent when a device needs to determine the MAC address that is associated with an IPv4 address. The device does not have an entry for that IPv4 address in its ARP table.

ARP request messages are broadcast to all devices within the local network segment using the following header information:

|                                                                                   |                                |                                                                                                                                                       |
|-----------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Destination MAC Address</b> | This is a broadcast address (FF-FF-FF-FF-FF-FF). Broadcast frames are accepted and processed by all devices on the LAN.                               |
|  | <b>Source MAC Address</b>      | This is the MAC address of the sender of the ARP request.                                                                                             |
|  | <b>Type</b>                    | ARP messages have a type field of 0x0806. This informs the receiving NIC that the data portion of the frame needs to be processed by the ARP process. |



### WHAT HAPPENS?

- The requesting device broadcasts the ARP request to all devices.
- All devices on the LAN receive the request.
- Only the device that owns the target IPv4 address will respond with an ARP reply.



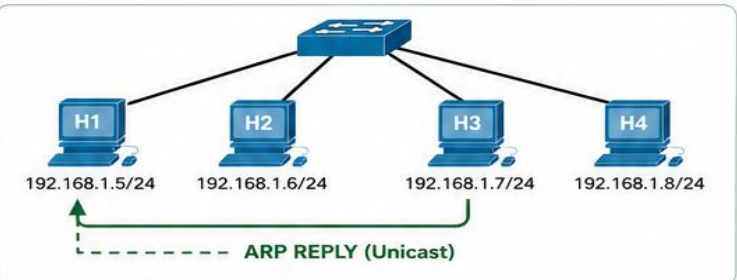
**NOTE:** ARP requests are broadcast within the local network segment. Routers do not forward broadcast ARP requests.

## 9.2.4 VIDEO – ARP OPERATION – ARP REPLY

Only the device that has the target IPv4 address will respond with an ARP reply. The ARP reply is sent as a unicast frame to the original requester.

The ARP reply is encapsulated in an Ethernet frame using the following header information:

|                                                                                     |                                |                                                                                                                                                    |
|-------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <b>Destination MAC Address</b> | This is the MAC address of the sender of the ARP request.                                                                                          |
|  | <b>Source MAC Address</b>      | This is the MAC address of the sender of the ARP reply (the device that owns the target IP).                                                       |
|  | <b>Type</b>                    | ARP messages have a type field of 0x0806. This informs the receiving NIC that the data portion of the frame needs to be passed to the ARP process. |



### AFTER RECEIVING THE ARP REPLY

- ✓ The requesting device adds the IPv4 and MAC address to its ARP table.
- ✓ Packets destined for that IPv4 address can now be sent using the corresponding MAC address.



### IMPORTANT POINTS

- ✓ If no device responds to the ARP request, the packet is dropped because a frame cannot be created.
- ✓ ARP table entries are time stamped. If a device does not receive a frame from that device before the timestamp expires, the entry is removed from the ARP table.
- ✓ Static ARP entries do not expire and must be manually removed.

### NOTE:

IPv6 uses a similar process called **ICMPv6 Neighbor Discovery (ND)**, which uses neighbor solicitation and neighbor advertisement messages, similar to IPv4 ARP requests and replies.

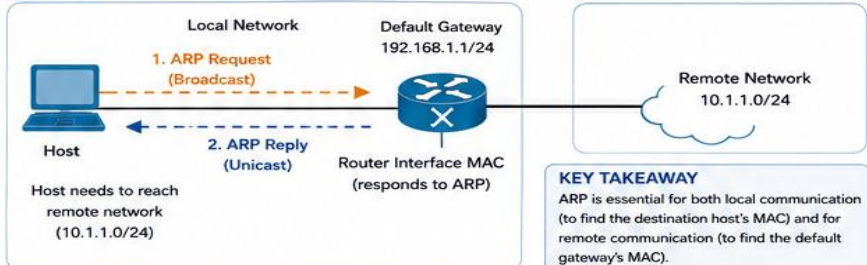
## 9.2.5 VIDEO – ARP ROLE IN REMOTE COMMUNICATIONS

When a host needs to send a packet to a remote network (IPv4 address not on the local network), it must send the frame to its default gateway (router interface).

Therefore, the host uses ARP to learn the MAC address of the default gateway's interface on the local LAN.

The ARP process is exactly the same: the host broadcasts an ARP request for the default gateway IP. The router interface (default gateway) responds with an ARP reply containing its MAC address.

Once the host has the gateway's MAC address, it builds the Ethernet frame with that MAC address as the destination and sends the packet to the gateway.



### QUICK SUMMARY



#### ARP Request

Broadcast to all devices on the local LAN to find the MAC of a target IP.



#### ARP Reply

Only the target device replies with its MAC address (unicast).



#### ARP Table

Maps IP ↔ MAC, stored in RAM. Entries time out.



#### Remote Communication

ARP is used to find the MAC of the default gateway.



#### IPv6 Equivalent

Uses ICMPv6 Neighbor Discovery (ND) instead of ARP.

# ARP (ADDRESS RESOLUTION PROTOCOL) – PART 2

## 9.2.6 REMOVING ENTRIES FROM AN ARP TABLE



ARP entries are removed from the ARP table if they have not been used for a period of time.

- The time depends on the operating system.
- Example (newer Windows): entries are stored between **15 and 45 seconds**.

PC A's ARP Cache (Example)

| IPv4 Address | MAC Address |
|--------------|-------------|
| 192.168.1.1  | 00:0D       |



An entry is removed if it is not used within 15–45 seconds.



Commands can be used to manually remove some or all entries from the ARP table. After an entry is removed, the device must perform ARP again to learn the mapping.

192.168.1.120/24  
MAC 00-0B



192.168.1.110/24  
MAC 00-0A



192.168.1.50/24  
MAC 00-0C



192.168.1.1/24  
MAC 00-0D



G0/0/0



Note: MAC addresses are shortened for demonstration purposes.

## 9.2.7 ARP TABLES ON NETWORKING DEVICES

ARP tables map IPv4 addresses to MAC addresses.

### On Cisco Router (IOS)

```
R1# show ip arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.10.1    -         a0e0.af0d.e140 ARPA   GigabitEthernet0/0/0
Internet 209.168.200.225 -         a0e0.af0d.e141 ARPA   GigabitEthernet0/0/1
Internet 209.165.200.226 1         a03d.6fe1.9d91 ARPA   GigabitEthernet0/0/1
R1#
```

### On Windows 10 PC

```
C:\Users\PC> arp -a
Interface: 192.168.1.124 --- 0x10
Internet Address      Physical Address  Type
192.168.1.1          c8-d7-19-cc-a8-86 dynamic
192.168.1.101        08-3e-0c-f5-f7-77 dynamic
192.168.1.110        08-3e-0c-f5-f7-56 dynamic
192.168.1.112        ac-b3-13-4a-bd-d0 dynamic
192.168.1.117        08-3e-0c-f5-f7-5c dynamic
192.168.1.126        24-77-03-45-5d-c4 dynamic
192.168.1.146        94-57-a5-0e-5b-02 dynamic
192.168.1.255        ff-ff-ff-ff-ff-ff static
224.0.0.22           01-00-5e-00-00-16 static
224.0.0.251          01-00-5e-00-00-fb static
239.255.255.250      01-00-5e-7f-ff-fa static
255.255.255.255      ff-ff-ff-ff-ff-ff static
C:\Users\PC>
```

## 9.2.8 ARP ISSUES – ARP BROADCASTS AND ARP SPOOFING



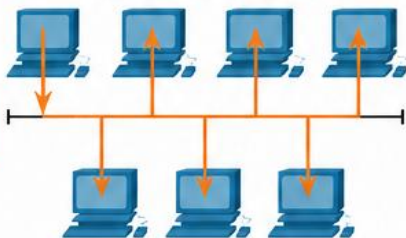
### ARP BROADCASTS

- ARP requests are broadcast frames.
- Every device on the local network receives and processes the broadcast.
- In a typical business network, impact is usually minimal.
- But in large networks with many devices, broadcasts can reduce performance for a short time.
- After devices reply, broadcasts stop and the impact is minimal.

All devices powered on at the same time

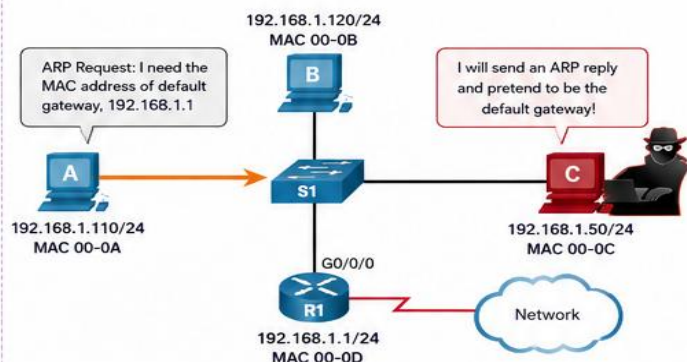
Shared Media (multiple access)

ARP broadcasts can flood the local media.



### ARP SPOOFING (ARP POISONING)

- A threat actor can send false ARP replies to associate their MAC address with another device's IPv4 address (e.g., the default gateway).
- Victim devices then send traffic to the wrong MAC address.
- The attacker can intercept, modify, or block the traffic.
- The attacker may also reply with its own IP's MAC address to maintain connectivity while eavesdropping.
- Enterprise switches use Dynamic ARP Inspection (DAI) as a mitigation technique.



Note: MAC addresses are shortened for demonstration purposes.

### KEY TAKEAWAYS



#### ARP Cache Entries

Removed if not used for 15–45 seconds (Windows example).



#### View ARP Table

- Cisco: `show ip arp`
- Windows: `arp -a`



#### ARP Broadcasts

Flood all devices on the LAN; usually minimal impact in small networks.



#### ARP Spoofing Risk

Attackers can send false ARP replies to intercept or disrupt traffic.



#### Mitigation

Use Dynamic ARP Inspection (DAI) on enterprise switches.

# IPV6 NEIGHBOR DISCOVERY (ND)

## 9.3.1 VIDEO – IPV6 NEIGHBOR DISCOVERY

- If your network uses IPv6, the Neighbor Discovery protocol, or ND, matches IPv6 addresses to MAC addresses.
- ND replaces ARP (used in IPv4).
- It also provides router discovery, address resolution and redirection services.

### Why ND is needed

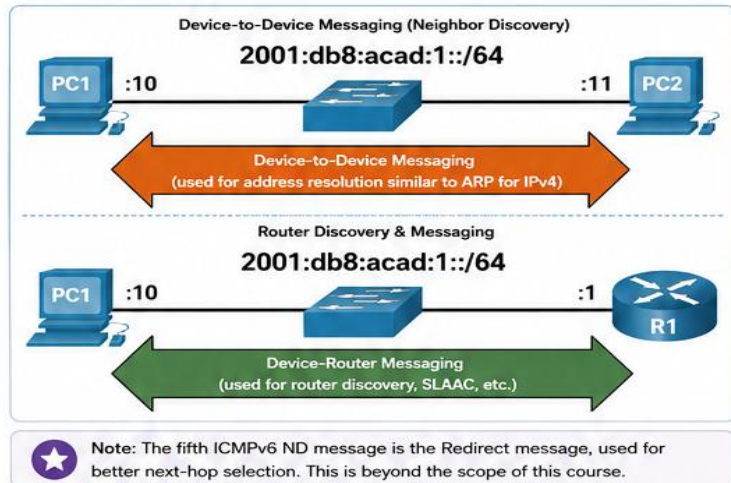
On an Ethernet network, an IPv6 device must know the MAC address of another device to send a frame. ND helps devices discover that MAC address from a known IPv6 address.

## 9.3.2 IPV6 NEIGHBOR DISCOVERY MESSAGES

IPv6 ND is sometimes called ND or NDP. It uses ICMPv6 messages to discover neighbors and routers and to provide other services.

ND uses the following ICMPv6 message types:

- 1 Neighbor Solicitation messages
- 2 Neighbor Advertisement messages
- 3 Router Solicitation messages
- 4 Router Advertisement messages
- 5 Redirect Message



Neighbor Solicitation and Neighbor Advertisement are used for device-to-device messaging such as address resolution (similar to ARP for IPv4). Devices include both host computers and routers.

Router Solicitation and Router Advertisement are for messaging between devices and routers. Typically router discovery is used for dynamic address allocation and stateless address autoconfiguration (SLAAC).

## 9.3.3 IPV6 NEIGHBOR DISCOVERY – ADDRESS RESOLUTION

- Like ARP in IPv4, IPv6 ND is used to find the MAC address of a device when its IPv6 address is known.
- ICMPv6 Neighbor Solicitation and Neighbor Advertisement messages perform MAC address resolution.
- Example: PC1 wants to ping PC2 at 2001:db8:acad:1::11. To find PC2's MAC address, PC1 sends an ICMPv6 Neighbor Solicitation message.



### How it works

- PC1 sends a Neighbor Solicitation (NS) message to find out who has IPv6 address ::11.
- Only the device with that IPv6 address (PC2) replies.
- PC2 sends a Neighbor Advertisement (NA) message back with its MAC address.
- PC1 now knows the MAC address of PC2 and can send the frame.



### Key Points

- NS and NA messages use special Ethernet and IPv6 multicast addresses.
- This lets the Ethernet NIC of the receiving device determine if the message is for it (or not) without sending it up for full processing.
- PC2 replies to the request with an NA message that includes its MAC address.

### QUICK SUMMARY



ND matches IPv6 addresses to MAC addresses (like ARP in IPv4).



ND uses 5 ICMPv6 message types: NS, NA, RS, RA, and Redirect.



Device-to-device messaging uses NS and NA (address resolution).



Router discovery and messaging use RS and RA (SLAAC, etc.).



ND makes IPv6 networks work efficiently without ARP.

### ND MESSAGE TYPES AT A GLANCE

| Message Type                | Purpose                                                          |
|-----------------------------|------------------------------------------------------------------|
| Neighbor Solicitation (NS)  | Ask for the MAC address of an IPv6 address (address resolution)  |
| Neighbor Advertisement (NA) | Reply with MAC address (or other info) to NS                     |
| Router Solicitation (RS)    | Host asks routers for information                                |
| Router Advertisement (RA)   | Router advertises configuration info (prefixes, default gateway) |
| Redirect Message            | Router tells host a better next-hop to use                       |

### IPV6 ND VS IPV4 ARP

| IPv4 (ARP)                           | IPv6 (ND)                                         |
|--------------------------------------|---------------------------------------------------|
| Maps IPv4 addresses to MAC addresses | Maps IPv6 addresses to MAC addresses              |
| Uses ARP request (broadcast)         | Uses ICMPv6 NS (multicast)                        |
| ARP reply (unicast)                  | ICMPv6 NA (unicast)                               |
| Separate router discovery methods    | ND includes router discovery (RS/RA) and more     |
| Limited functions                    | More functions (SLAAC, redirects, neighbor cache) |

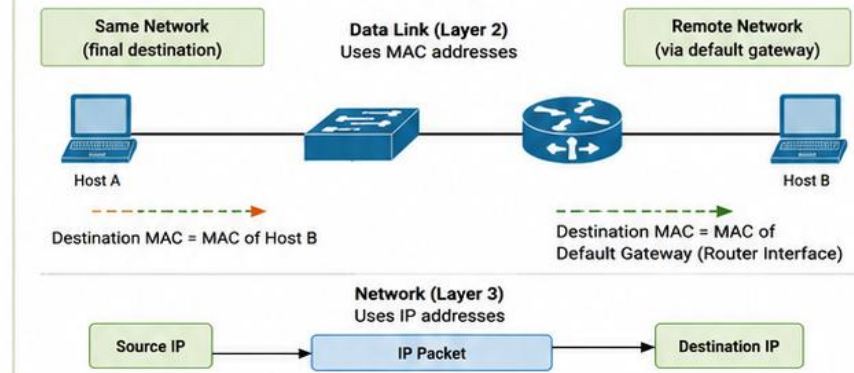
# 9.4.1 WHAT DID I LEARN IN THIS MODULE?

## SUMMARY OF KEY CONCEPTS

### 1 MAC and IP

- Layer 2 physical addresses (Ethernet MAC addresses) are used to deliver a frame on the local network.
- The IP packet is encapsulated in a frame using the MAC addresses.
- If the destination IP is on the **same network**, the destination MAC is the final device.
- If the destination IP is on a **remote network**, the destination MAC is the **default gateway** (router interface).
- Along each link, the destination MAC is the MAC of the **next-hop** device on that link.
- IPv4 uses **ARP** to map IP addresses to MAC addresses.
- IPv6 uses **ICMPv6 Neighbor Discovery (ND)** to perform this mapping.

#### How a Frame is Delivered



**Key Point:** MAC addresses change at every hop; IP addresses remain end-to-end.

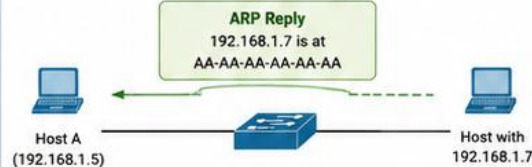
### 2 ARP (Address Resolution Protocol)

- Purpose:** Maps IPv4 addresses to MAC addresses on a local network.
- Uses an ARP table (cache) to store temporary IP-to-MAC mappings.
- If no entry exists, the device sends an **ARP Request (broadcast)**.
- Only the device with the target IP replies with an **ARP Reply (unicast)**.
- ARP entries have a timer and expire after a period of inactivity. Entries can be removed manually with commands.
- Security:** ARP broadcasts can flood the network; **ARP spoofing (poisoning)** can redirect traffic to a wrong MAC address.

#### ARP Request (Broadcast)



#### ARP Reply (Unicast)



#### Example ARP Table

| IPv4 Address | MAC Address       |
|--------------|-------------------|
| 192.168.1.1  | 00:0D:AA:AA:AA:AA |
| 192.168.1.7  | 00:0D:BB:BB:BB:BB |
| ...          | ...               |

#### ARP Uses (on local network)

- Address resolution (IP → MAC)
- Next-hop determination (default gateway MAC)
- Interface of local router

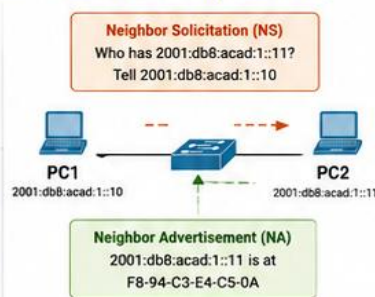
#### Note

ARP is for IPv4 only. IPv6 uses ICMPv6 ND.

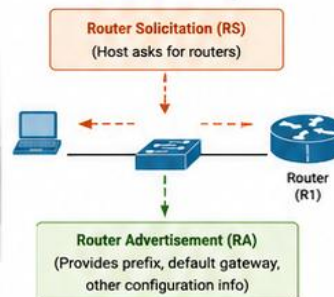
### 3 Neighbor Discovery (IPv6 ND)

- IPv6 does not use ARP. It uses Neighbor Discovery (ND).
- Built on **ICMPv6** messages to provide: address resolution, router discovery, and redirection services.
- ND Message Types (ICMPv6):**
  - Neighbor Solicitation (NS)
  - Neighbor Advertisement (NA)
  - Router Solicitation (RS)
  - Router Advertisement (RA)
  - Redirect Message

#### Address Resolution (similar to ARP)



#### Router Discovery



#### ND Uses

- Address resolution (IPv6 → MAC)
- Router discovery
- Configuration (prefixes, default gateway, etc.)
- Redirection (better next-hop selection)

ND messages use special Ethernet (multicast) and IPv6 multicast addresses so devices can discover neighbors without flooding the entire network.

## AT A GLANCE – KEY DIFFERENCES

| Feature       | ARP (IPv4)                                    | Neighbor Discovery (IPv6 ND)                                     |
|---------------|-----------------------------------------------|------------------------------------------------------------------|
| Purpose       | Maps IPv4 addresses to MAC addresses          | Maps IPv6 addresses to MAC addresses + router discovery & more   |
| Protocol      | ARP (Layer 2)                                 | ICMPv6                                                           |
| Messages      | ARP Request (broadcast), ARP Reply (unicast)  | NS, NA, RS, RA, Redirect (ICMPv6 messages)                       |
| Use           | Address resolution, next-hop (gateway) lookup | Address resolution, router discovery, configuration, redirection |
| Security Risk | ARP spoofing (poisoning)                      | Threats exist, mitigated with features like Secure ND (SEND)     |